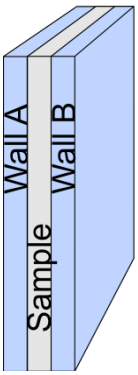
TeraLyzer QuickStart Multilayer

Lytera UG (haftungsbeschränkt)

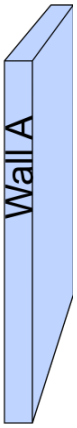
Revision 1.0

Before using the TeraLyzer Software Suite for multilayer analysis with your own data, we highly recommend to go through the multilayer data analysis with the sample datasets we provide. The data was measured with a TDS setup using photoconductive antennas both as emitter and as detector.

In this tutorial, we assume that you would like to investigate a liquid sample within a glass cuvette. To perform a multilayer analysis, the material parameters and thicknesses of all layers except the sample layer have to be determined in advance.



In our case, the cuvette consists of two glass windows with the liquid sample enclosed in between. It is important to measure the material parameters of the single layer cuvette walls with high precision as any error in these measurements would directly induce inaccuracies when the multilayer system is considered.



Therefore, in a first step, each wall is separately measured. We recommend obtaining at least two references (Air) and two measurements of the wall. This procedure allows us to specify narrow error bars on the material parameters of the sample and ensures a high accuracy for the multilayer evaluation.

After measuring the walls, the single layer data extraction can be performed for each wall, following the steps described in section 1 of this document (see also main manual chapter 2.1), yielding the complex material parameters and the wall thickness.

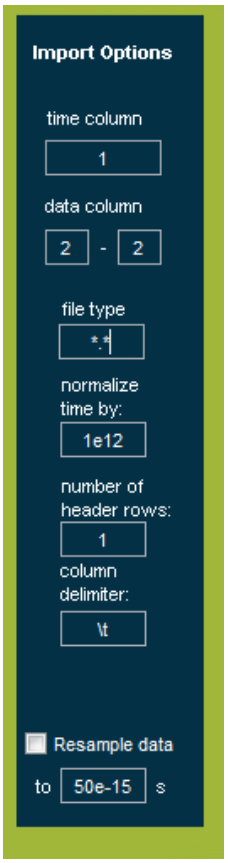
After all layers except the sample layer have been characterized, the multilayer sample is placed in the terahertz system and again, reference (Air) and sample measurements are performed followed by the material parameter extraction procedure described in section 2 of this document (see also main manual chapter 2.2).

1 Data Extracion of the Walls

The files *Glas_700µm.txt* and Glas_*700µm2.txt* are measured THz datasets of a 700 µm thick piece of glass. The corresponding reference time domain waveforms, measured without a sample in the THz beam path, are provided in the files *Reference_Glas.txt* and *Reference_Glas2.txt*. The first column of the file holds the time axis in picoseconds and the second one the THz time domain waveform.

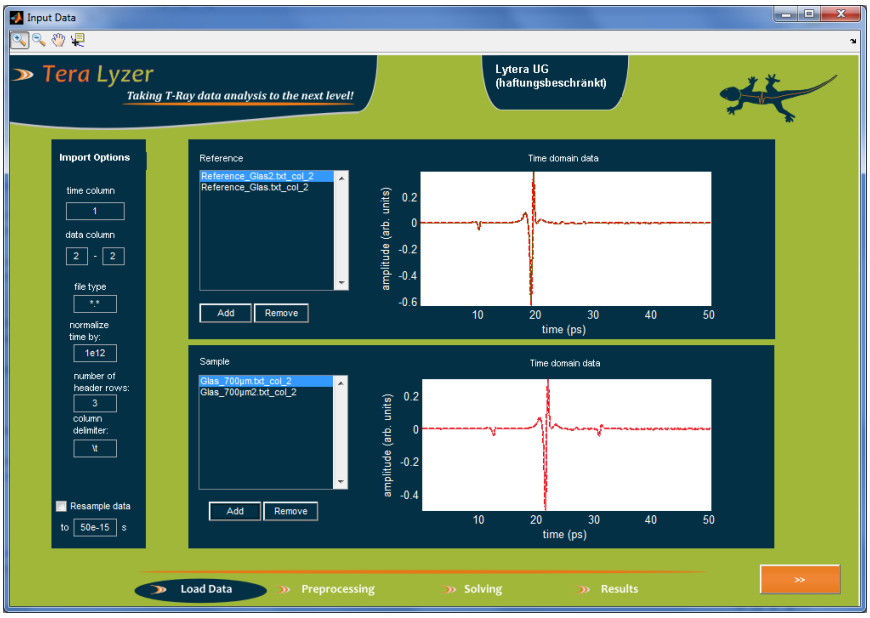
```
Time                Data
0.000000000000000   0.001953140000000
0.080055382847556   0.002014180000000
0.160110765695113   0.002014180000000
0.240166148542669   0.001953140000000
0.320221531390226   0.002014180000000
0.400276914237782   0.002014180000000
0.480332297085339   0.002014180000000
0.560387679932895   0.001953140000000
```

To load a data file into the TeraLyzer, it is important to choose the right import settings before adding the file. In this case, the first column is the time column and the data is stored in column two. We want to load data with any file extension, so the field *file type* contains a wildcard *.*.



In this example we have one header row and the delimiter is a tab-stop. We don't want to resample the data, so we leave the resample box unchecked.

After the specification of the input settings, the reference files can be loaded by clicking on the *Add* button. After navigating to the folder, where the data is stored, multiple files can be added at once.

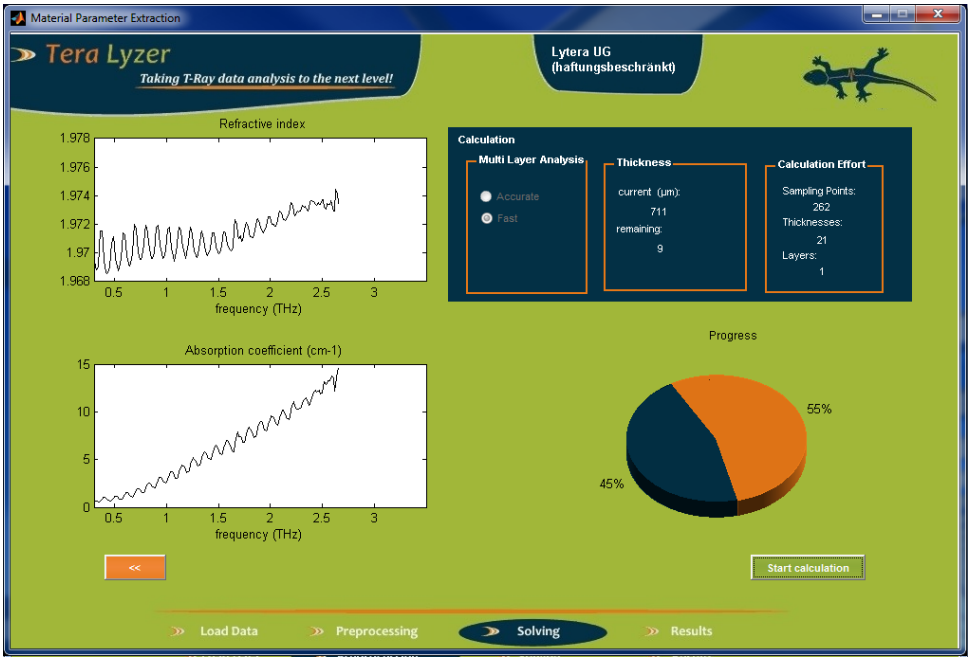


A preview of the pulse is shown, once the data is added. After selecting the sample data as well, one can proceed by clicking the >> button.

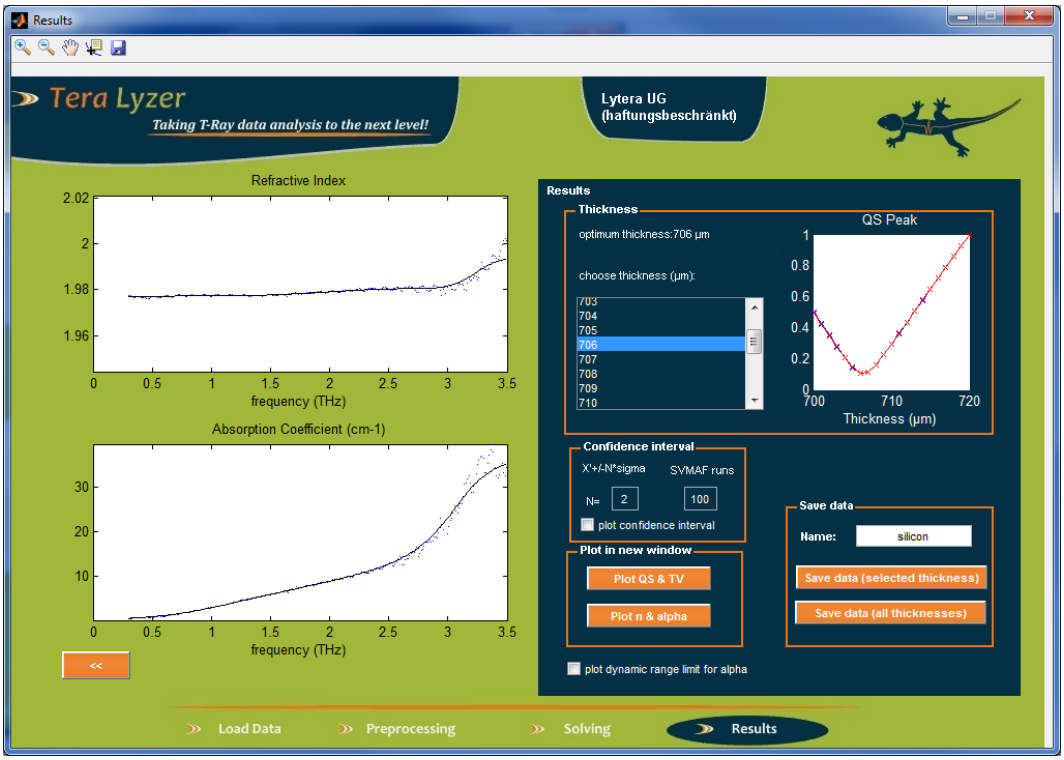
In the next window, several settings regarding the signal processing and the sample details can be chosen. For the sample data, we provide a settings file (*parset_Glas_700µm.mat*), which can be loaded by clicking the open settings button found in the top menu.

The signal processing settings have to be applied to the data by clicking the *Apply* button. The approximate thickness of the sample as well as the frequency range of interest has to of

In the next window, some information about the calculation effort is provided. The algorithm has to be started by clicking the *Start calculation* button. The calculation of the material parameters may take some time.



The results of the calculation are presented in the last window. The sample's refractive index and absorption coefficient corresponding to the numerically determined sample thickness are shown.



In the last step, choose a filename like *WallA* and save the data by clicking the *Save data (selected thickness)* button.

2 Data Extraction of the Liquid Sample Layer

To proceed with this step, you need the material parameter files obtained by evaluating the cuvette walls as described in section 1 of this document.

The extraction process of the sample layer material parameter information begins with similar steps as the previously discussed single layer extraction. In the first window, select the reference and the sample files, corresponding to measurements of air and the filled cuvette. As before, proceed to the next window by clicking the *Next* (>>) button. In the second window use the same preprocessing settings as with the single layer files and click on *Apply.* Insert the assumed thickness of the sample layer thickness and modify the search interval. Be aware that the multilayer algorithm is much more time

consuming than the single layer algorithm, so starting with 3-5 thicknesses is a good choice here. To setup the multilayer properties, click on *Multilayer setup*, which will open the multilayer setup window:

In the next step, the material parameters of the walls have to be loaded. To do so, click on *Get transmission data file* and load the material parameters you saved before as *WallA.* Click on *Add layer* to add the cuvette wall to the layer stack. Afterwards, add the sample layer by clicking the *Add investigated layer* button. At last, load the material parameters for *Wall B* and add the layer. We highly recommend using the SVMAF values for the calculation unless the material parameters of the previously characterized layers exhibit sharp spectral features (usually not the case).

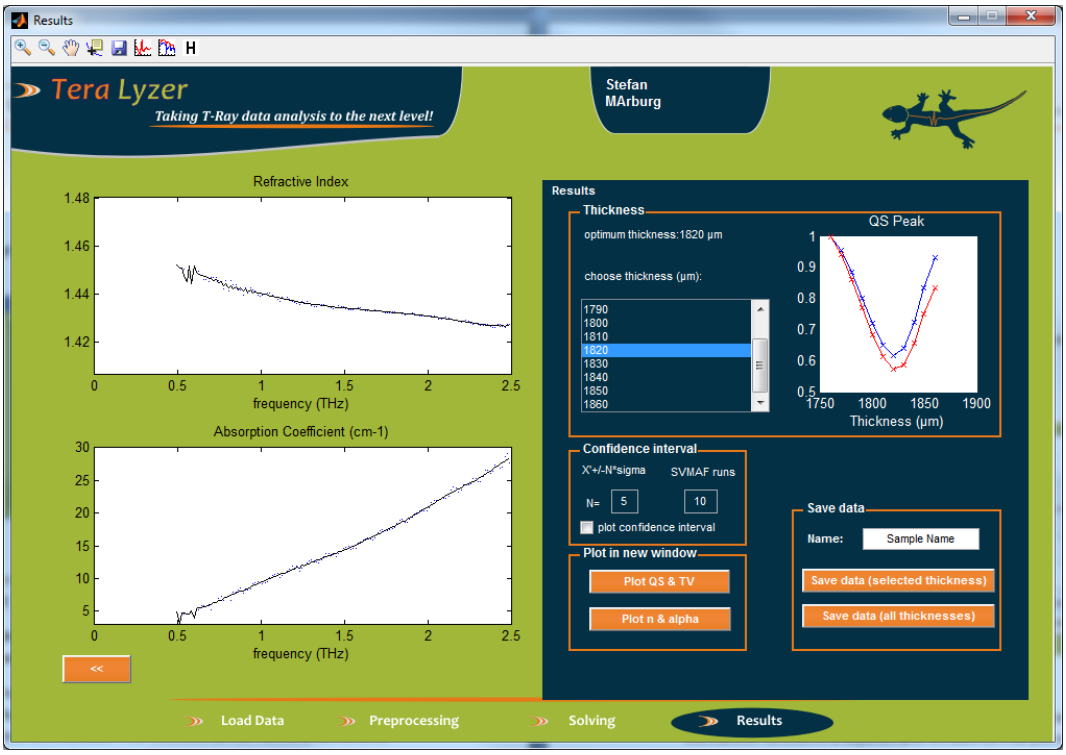
Please note that the order of the layers as displayed in the list box from bottom to top has to be the order that the incoming waves encounters the layers, as indicated by the white arrow to the right.

When you finished building the layer stack, proceed with clicking *OK.*

In the next window, click on *Start calculation* button.

The TeraLyzer software offers two different approaches to create a theoretical multilayer transfer function: A Transfer Matrix and a Ray Tracing formulation. While the transfer matrix approach offers a much higher computational efficiency, it inherently assumes that all internal reflections fall within the measured time window. In contrast to that, the Ray Tracing approach accounts for the finite time window but is relatively complex and computation intensive.

For more information on the algorithms, please refer to our main manual, 2.2



After the calculations are finished, the results window with the quasi space curves and the material parameters appears. As in the case of the single layer evaluation, the correct thickness of the layer between the cuvette walls is found at the minimum of the V-shaped quasi space curves. More information on how to correctly set the confidence interval parameters is provided in the main manual, 3.4

After the confidence interval settings and the sample name are chosen, click on the *Save data* button to save your results.

Congratulations! You just finished the multilayer data extraction! Now, it is time to try the procedure with your own data!

Thanks for using the TeraLyzer! Enjoy!

Sincerely Yours,

The Lytera Team